

Ali Raza

☎ +92 300 7651218 ✉ mirzaalirazafsd@gmail.com 🔗 ali-raza-0a1282281 🎧 alyrraza 🏆 kaggle/f223399mirzaaliraza

SUMMARY

AI/ML Engineer with **3+** years hands-on experience in **Deep Learning**, **Computer Vision**, and **LLM**-powered systems. Strong foundation in PyTorch, TensorFlow, model training and evaluation, API integration, and production deployment. Experienced with **LangChain**, **LangGraph**, and YOLO-based pipelines.

TECHNICAL SKILLS

ML / DL & CV: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, **Optimization**(LoRA, QLoRA, PEFT) , **ML Models**(XGBoost, LightGBM), fine-tuning, transfer learning, **Computer Vision**(YOLOv8, TransU-Net, U-Net, ViT, SAM, ResNet, DenseNet, BioViL-T, GradCAM, CBAM, OpenCV), **Deep Generative Models**(GANs, VAEs, Probabilistic Models, Diffusion, Mamba)

NLP, LLMs & Data: BERT, LlamaIndex, LangGraph, RAG pipelines, LangChain, FAISS, Pinecone, embeddings, prompt engineering, OpenAI API, Gemini API, Anthropic API, Pandas, NumPy, SQL, PostgreSQL, MongoDB

MLOps, Deployment & Programming: MLflow, Docker, Kubernetes, KubeFlow, Prometheus, Grafana, Evidently AI, FastAPI, CI/CD, GitHub Actions, AWS EC2, inference optimization, GPU deployment, Python, C++, JavaScript, Git

EXPERIENCE

ML Engineer — Freelance | International Clients

2023 – Present

Remote, Faisalabad, Pakistan

- Designed **end-to-end ML pipelines** — preprocessing, training, evaluation, and **scalable production deployment** via FastAPI + Docker with real-time **model performance monitoring**.
- Built and deployed **YOLOv8 computer vision system** — inference-optimized REST API, containerized with Docker, GPU deployment on Vast.ai, integrated with React frontend on Vercel.
- Fine-tuned **LLMs (LoRA/QLoRA)** via HuggingFace PEFT; built **RAG pipelines** with LangChain, FAISS, and Pinecone for document intelligence systems.
- Maintained **MLflow** experiment tracking and model versioning; automated deployments via **CI/CD** pipelines using GitHub Actions.

Data Analyst Intern | TAKMIL

2024

Louisville, Kentucky, USA · Remote

- Cleaned and analyzed large-scale datasets via Python (Pandas, NumPy, SQL); built **Power BI dashboards** — proposed optimizations improving workflow efficiency by **30%**.

RESEARCH & KEY PROJECTS

RadGuard: Multimodal ML System for Medical Imaging — Final Year Thesis

2025–2026

BioViL-T · CXR-BERT · PyTorch · MLflow · Docker · Kubernetes · KubeFlow · FastAPI · AWS EC2

- Built multimodal deep learning system detecting 4 error types in AI-generated chest X-ray reports across 14 conditions on 74,060 MIMIC-CXR samples (**Val F1=0.66**); discovered and fixed 10 critical labeling bugs producing 147K false labels.
- Deployed on **AWS EC2** with Docker + Kubernetes autoscaling, **KubeFlow** retraining pipeline, Prometheus+Grafana monitoring, A/B shadow testing, and **FAISS RAG** chatbot for clinical querying.

ProgressGAN-CXR — Generative AI Research

2025–2026

PyTorch · DCGAN · WGAN-GP · Knowledge Distillation · FastAPI · React · HuggingFace · Docker

- Trained 4 severity-conditioned GAN architectures for continuous COVID-19 disease progression simulation on 21,165 chest X-rays — proposed novel 3-metric evaluation framework (FID, Spearman r, SSIM) showing FID alone is insufficient for medical GANs.
- Deployed full-stack production application — FastAPI inference server on HuggingFace Spaces with Prometheus monitoring, React + TypeScript frontend on Vercel, model weights on HF Hub; live demo at progressgan-cxr.vercel.app.

Segmentation & Classification of CIEDs — ACR 2026

2024–2025

TransU-Net · SAM · DINO · ViT · nnU-Net · U-Net

- Designed **TransU-Net** combining transformer attention and U-Net skip connections for automated pacemaker/ICD detection and classification in chest X-rays — fully automatic localization-to-classification pipeline.

CV Interview Coach — AI Multi-Agent System

2025

LangGraph · LangChain · Groq · ChromaDB · LlamaIndex · Whisper · LiveKit · FastAPI · Next.js

- Built a **3-agent pipeline** with Supervisor MCP orchestrator — CV/JD analyzer, RAG-augmented question generator, and real-time voice interview agent with **self-healing retry loops** (critique injection, max 3 retries).

EDUCATION

BSc Computer Science | **FAST-NUCES, Faisalabad**

2022 – 2026

Co-Head Hackathon ACM Daira 2025 | TAKMIL Data Science Internship 2024

PUBLICATIONS

Ali Raza et al. “Segmentation & Classification of CIEDs in Chest X-rays” — ACR 2026